SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry based Problem-Solving

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4-AT2**	**Assessment:**	AT-2-Industry Based Problem Solving
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	Analyze datasets using descriptive statistical measures such as mean, variance, sampling methods, covariance, and correlation to identify patterns, relationships, and support data-driven decision-making.		
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Section / Batch:	BTech AI&DS	Date:	20-08-2026

Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

INDUSTRY BASED PROBLEM-SOLVING

Confidence Interval and Hypothesis Testing Using Confidence Interval

1. Software Industry — Confidence Interval

Problem-Solving Question:

A software company measures the response time of 100 API requests. The sample mean is 240 ms and standard deviation is 35 ms. Construct a 95% confidence interval for the average API response time.

Solution:

Given: $n = 100$, $\bar{x} = 240$ ms, $s = 35$ ms.

Since the population standard deviation is not given, use the t-distribution with df = 99. For 95% confidence, $t_{0.025,99} \approx 1.984$.

Standard Error = $s/\sqrt{n} = 35/\sqrt{100}$ = 3.5 ms.

Margin of Error = $t \times SE = 1.984 \times 3.5$ = 6.944 ms.

95% Confidence Interval = 240 ± 6.944.

Therefore, 95% CI = (233.056 ms, 246.944 ms) ≈ (233.06 ms, 246.94 ms).

Interpretation: We are 95% confident that the population average API response time lies between approximately 233.06 ms and 246.94 ms.

2. E-commerce — Confidence Interval

Problem-Solving Question:

An e-commerce platform finds that 420 out of 600 randomly selected visitors purchased a product. Construct a 95% confidence interval for the true conversion rate.

Solution:

Given: $x = 420$ purchases and $n = 600$ visitors.

Sample conversion rate $\hat{p} = 420/600$ = 0.70 = 70%.

Standard Error = $\sqrt{[\hat{p}(1-\hat{p})/n]} = \sqrt{[(0.70)(0.30)/600]}$ ≈ 0.01871.

For a 95% confidence interval, $z = 1.96$.

Margin of Error = 1.96×0.01871 ≈ 0.03667.

95% Confidence Interval = 0.70 ± 0.03667.

Therefore, 95% CI = (0.6633, 0.7367) ≈ (66.33%, 73.67%).

Interpretation: We are 95% confident that the true conversion rate is between approximately 66.33% and 73.67%.

3. Telecom — Confidence Interval

Problem-Solving Question:

A telecom company wants to estimate the average monthly data consumption of its users. A sample of 225 users has a mean consumption of 14.8 GB and standard deviation of 4.5 GB. Construct a 99% confidence interval for the population mean.

Solution:

Given: $n = 225$, $\bar{x} = 14.8$ GB, $s = 4.5$ GB.

Since the population standard deviation is not given, use the t-distribution with df = 224. For 99% confidence, $t_{0.005,224} \approx 2.601$.

Standard Error = 4.5/√225 = 4.5/15 = 0.30 GB.

Margin of Error = 2.601 × 0.30 = 0.7803 GB.

99% Confidence Interval = 14.8 ± 0.7803.

Therefore, 99% CI = (14.0197 GB, 15.5803 GB) ≈ (14.02 GB, 15.58 GB).

Interpretation: We are 99% confident that the population average monthly data consumption lies between approximately 14.02 GB and 15.58 GB.

4. Manufacturing — Hypothesis Testing Using CI

Problem-Solving Question:

A machine historically produces components with a mean diameter of 50 mm. A sample of 36 components has a mean diameter of 50.8 mm and standard deviation of 2.4 mm. Using a 95% confidence interval, determine whether 50 mm is a plausible population mean.

Solution:

Given: $n = 36$, $\bar{x} = 50.8$ mm, $s = 2.4$ mm.

Use the t-distribution with df = 35. For 95% confidence, $t_{0.025,35} \approx 2.030$.

Standard Error = 2.4/√36 = 0.40 mm.

Margin of Error = 2.030 × 0.40 = 0.812 mm.

95% Confidence Interval = 50.8 ± 0.812 = (49.988 mm, 51.612 mm).

Because 50.00 mm lies within the confidence interval (49.988 mm, 51.612 mm), **50 mm is a plausible population mean at the 95% confidence level.**

Conclusion: 50 mm is a plausible population mean at the 95% confidence level. The sample does not provide sufficient evidence to reject the historical mean of 50 mm using this confidence-interval approach.

5. Banking — Hypothesis Testing Using CI

Problem-Solving Question:

A bank claims that its average loan-processing time is 5 days. A sample of 49 applications has an average processing time of 5.6 days with standard deviation 1.4 days. Construct a 95% confidence interval and determine whether the company's claim is reasonable.

Solution:

Given: n = 49, $\bar{x}$ = 5.6 days, s = 1.4 days.

Use the t-distribution with df = 48. For 95% confidence, $t_{0.025,48} \approx 2.011$.

Standard Error = 1.4/√49 = 0.20 days.

Margin of Error = 2.011 × 0.20 = 0.4022 days.

95% Confidence Interval = 5.6 ± 0.4022 = (5.1978 days, 6.0022 days) ≈ (5.20 days, 6.00 days).

The claimed population mean of 5 days is inside the 95% confidence interval.

Conclusion: At the 95% confidence level, 5 days is a plausible population mean. Therefore, the sample does not provide sufficient evidence to reject the bank's claim.